

CS-3083 Project 3 Report

Bradley Titagwan

Task 2: Naive Bayes Text Classification Report

Overview

In this task, a Multinomial Naive Bayes (MNBC) classifier was implemented to perform text classification on two datasets. The first dataset was the provided academic dataset consisting of four classes: biology, economics, physics, and world history. The second dataset was an external dataset used for additional evaluation, where the first 1200 samples were used for training and the remaining samples were used for testing.

The objective of this task was to evaluate how preprocessing techniques and smoothing parameters affect classification performance and to analyze the strengths and limitations of the Naive Bayes model.

Methodology

The Multinomial Naive Bayes model was used, which calculates:

- The prior probability of each class
- The likelihood of each word given a class

To prevent numerical underflow, log probabilities were used when computing class scores. Laplace smoothing was applied to handle unseen words and prevent zero probabilities.

The model assumes that all words in a document are conditionally independent given the class. While this simplifies computation and makes the model efficient, it limits the model's ability to capture context and relationships between words.

Preprocessing

Several preprocessing techniques were applied to improve model performance:

- Conversion of text to lowercase
- Removal of punctuation
- Removal of stopwords
- Filtering of short tokens
- Lemmatization to reduce words to their base forms

Multiple preprocessing configurations were tested to determine their impact on accuracy.

Results – Provided Dataset

The provided dataset consisted of four classes: biology, economics, physics, and world history.

The following preprocessing configurations were evaluated:

lower_only → 0.855

lower_punct → 0.855

lower_punct_stop → 0.920

lower_punct_stop_alpha05 → 0.920

lower_punct_stop_alpha15 → 0.925

The best performance was achieved using lowercase conversion, punctuation removal, stopword removal, and a smoothing parameter of $\alpha = 1.5$, resulting in an accuracy of 93%.

Confusion Matrix Analysis

The confusion matrix indicates that most predictions were correct, with relatively few misclassifications. The most common errors occurred between:

- Biology and physics
- World history and economics

These misclassifications are due to overlapping vocabulary between classes. Biology and physics share general scientific terminology such as “energy” and “system,” while world history and economics share terms such as “market,” “trade,” and “economy.”

Per-class performance showed strong precision and recall across all categories, with slightly lower recall for biology due to confusion with physics.

Misclassification Analysis

Misclassified examples provide insight into the limitations of the model:

- Biology sentences were sometimes classified as physics due to shared scientific vocabulary
- History sentences were sometimes classified as economics due to economic-related language

- Short sentences were more likely to be misclassified because they contained fewer distinguishing features

These results highlight that the model relies heavily on word frequency and cannot fully interpret contextual meaning.

Results – Extra Dataset

The external dataset was evaluated using a fixed split, where the first 1200 samples were used for training and the remaining samples were used for testing.

The model achieved high accuracy on this dataset, with performance varying slightly depending on preprocessing settings. Overall, the accuracy was higher than that of the provided dataset.

Analysis of Extra Dataset

The improved performance on the external dataset can be attributed to:

- Simpler class structure
- More distinct vocabulary between classes
- Strong keyword indicators that clearly distinguish categories

Preprocessing had a smaller impact on this dataset compared to the provided dataset, suggesting that the classification task was inherently easier.

Comparison of Datasets

The provided dataset is more complex due to:

- Multiple classes
- Greater overlap in vocabulary
- More nuanced and structured text

In contrast, the external dataset is easier to classify due to clearer separation between classes and more distinct word usage.

Stopword removal had the greatest impact on the provided dataset, while smoothing had a moderate effect. On the external dataset, preprocessing changes had a smaller effect on performance.

Additionally, reducing vocabulary size through preprocessing improved computational efficiency and reduced runtime.

Conclusion

The Multinomial Naive Bayes classifier achieved strong performance on both datasets. It performed best on the external dataset due to simpler classification structure and more distinct vocabulary, while still achieving high accuracy on the more complex provided dataset.

Preprocessing techniques, particularly stopword removal and smoothing, significantly improved performance on the provided dataset. However, the model remains limited by its independence assumption and inability to fully capture context.

Despite these limitations, the Naive Bayes classifier is an effective, efficient, and practical approach for text classification, especially when combined with appropriate preprocessing techniques.